

TOH YUN NING

Data Science Graduate — Machine Learning, Data Engineering & Semiconductor Manufacturing Analytics
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EDUCATION

Tunku Abdul Rahman University of Management and Technology (TAR UMT), Penang

Jul 2024 – Jul 2026

Bachelor of Data Science (Honours)

CGPA 3.8 with President's List (4.00) & Dean's List (3 trimesters), focused on Data Analysis & Computer Vision

Tunku Abdul Rahman University of Management and Technology (TAR UMT), Penang

Jul 2022 – Jun 2024

Diploma in Computer Science

CGPA 3.9, focused on programming fundamentals, data science basics and statistics

INTERNSHIP EXPERIENCE

Gospower Electric Technology Co., Ltd., Dongguan, China

Jan 2026 – Jul 2026

Research and Development Department Intern

- Rotated across a server power-supply (CRPS) manufacturing pipeline — production, SMT/AOI, EMC/reliability testing, automation, and QC — gaining ground-level exposure to manufacturing defect-tracking systems.
- Ran automated test campaigns on a Chroma 8000 system (11 test types incl. surge current, OCP, dynamic load).
- Built C# test-automation software (Windows Forms + NI-VISA) for the in-house ATS; completed STM32 firmware exercises (interrupts, PWM, ADC/DMA).
- Gained direct exposure to yield/defect-tracking pipelines in high-volume electronics manufacturing.

ViTrox Technologies Sdn Bhd, Batu Kawan, Penang, Malaysia

Oct 2023 – Feb 2024

Research and Development Department Intern

- Labelled and trained PyTorch-based image segmentation on PCB defect datasets to support automated X-ray inspection (AXI) R&D.
 - Produced technical evaluation reports benchmarking model performance.
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FINAL YEAR PROJECT

Void Segmentation in Land Grid Array (LGA) Components Using DeepLabV3+ and SegNeXt

with data sponsored by ViTrox Technologies Sdn Bhd

- Designed and compared modified SegNeXt and DeepLabV3+ semantic segmentation models to detect solder voids on PCB X-ray images.
 - Achieved 97.0% accuracy and 88.1% mean IoU with the modified SegNeXt, outperforming the modified DeepLabV3+ (OHM) at 94.9% / 82.5%.
 - Created an automated visual defect inspection web application using Streamlit.
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NOTABLE ASSIGNMENTS

Malay Lexicon Streaming Pipeline — Data Engineering

- Built a pipeline scraping Malay news via Kafka into Hadoop/PySpark, constructing a linguistic lexicon enriched with dictionary and Wikipedia data.
- Stored results in Neo4j with Redis caching; automated scheduling via cron — full pipeline from collection to storage.

Constellation Schema & Reporting — Data Warehouse Technology

- Designed a constellation schema with shared dimensions across multiple fact tables for a bus-operations dataset.
- Wrote PL/SQL cursor-driven reports for revenue growth, route profitability, and company performance ranking.

BERT Sentiment Classification (Yelp Reviews) — Artificial Intelligence

- Fine-tuned BERT for restaurant review sentiment classification across 3 iterations to resolve overfitting/underfitting — 97.1% accuracy, 96.2% macro F1; deployed a Streamlit demo UI for the model.
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TECHNICAL SKILLS

Languages: Python, SQL (Oracle PL/SQL, MySQL), R, Java, C, C++, JavaScript, HTML/CSS, SAS

ML/AI: PyTorch, scikit-learn, BERT/Transformers, computer vision (segmentation, watermarking), NLP (Llama)

Data Engineering: Kafka, Spark, Hadoop/MapReduce, ETL, data warehousing (star/constellation schema)

Databases & Tools: Oracle, MySQL, Neo4j, Redis, Power BI, SAS Viya, Git, Linux, Jupyter, Google Colab